SDGP 2024/25 CW 1
"Design and Documentation"
Template and Guidelines

Deadline
Thursday, 21st of November 2024, on or before 1 PM via
Blackboard and Google Form -
https://forms.gle/yzYKsyYwzV5MmJco8

Vivas will be from the 25th of November 2024 onwards. Viva's schedule will be shared later.

Other useful documents

- SDGP CW 1 Marking Scheme
- SDGP CW 1 Explanation Lecture
- Former IIT students FYP Sample Thesis (I have explained in the above lecture what this can be used for)

SDGP Module information for the Cover page

Module Name - Software Development Group Project
Module Code - 5COSC021C
Module Leader - Banuka Athuraliya

All Students must submit **two reports** as given below.

1. Group Report (Everyone should contribute to this report, and progress will be checked via Google Document history. Any student without the Google Document history will be awarded 0 marks.). The group report will contain the following chapters.
    - **Chapter 1: Introduction**
    - **Chapter 2: Existing work**
    - **Chapter 3: Methodology**
2. Individual Report. (Google document history will also be checked for this document to identify plagiarism). The individual report will contain the following chapters.
    - **Chapter 4: System Requirements Specification (SRS)**
    - **Chapter 5: Social, Legal, Ethical and Professional Issues**
    - **Chapter 6: System Architecture & Design**

<u>Guidelines to be covered for both reports</u>

The title of the project reports must be as follows -

“UoWID_IITID_Name - SDGPGROUPNO-CW 1 Group”

“UoWID_IITID_Name - SDGPGROUPNO-CW 1 Individual”

Ensure to include a good cover page with all the important information according to the university standards. The cover page should **not** contain a page number. Numbering starts from the declaration page and should be numbered using lowercase roman letters.

The main content of the document starts from this point onwards.

The main content should use an Indo-Arabic numbering system. There is no set page limit for this report.

You get marks for formatting. Therefore, make sure that the document is well-formatted. The following are mandatory for every team report,

- The top right of the document (header) should mention what chapter it is. Eg - "Chapter 1 - Introduction" or "Chapter 2 - Existing Work". It should change from chapter to chapter. This can be done using Section breaks. Watch some tutorials if you need to learn how to do it.
- Bottom right, you should have the page number
- Bottom left, you should have your team name and number (Eg - "IIT - SE-1")
- Use the appropriate font sizes (H1 size 16, H2 size 14, H3 size 12, and Body size 12) - and the font must be Times New Roman and NOT any other fancy fonts!
- Make sure the text is justified and proper alignment is there.
- Images and tables should have captions and be linked to the list of figures and tables.
- Don't have blank spacing unnecessarily.

It is a **must** to have a table of contents, a list of figures, and a list of tables. (All this must be auto-generated and NOT typed)

ChatGPT, Bard, Bing AI or any other AI-generated content will be penalized if found.

Plagiarism will be checked using Turnitin via BB, and Plagiarism will result in 0 marks.

Make sure to use uniform numbering in the document. Chapter 1 numbering should be 1, 1.1, 1.1.1, etc. Chapter 3 should be 3, 3.1, 3.2, 3.2.1, etc. Attention to detail matters a lot.

Have references to prove your statements or points. It will give credibility to what you say with references most of the time. Rather than you saying something, it is best to use valid documentation work to prove your statements. It is Mandatory to use the Harvard referencing style.

Bonus marks will be given if the student or team has won any **recognized** competition. (Individual or group). The more extra work you do, the better it is for your CV.

It is recommended that you use Google docs to collaborate with your team (as this will be checked during the viva to see the Individual contribution)

The report must be written in the third person; therefore, please don't use terms like "We" or "I". Academic English is recommended for this report.

Section 1 - Topics to be covered as a group report

The following topics must be covered in the report.

1. Cover Page
2. Declaration Page
3. Abstract + Keywords
4. Acknowledgement
5. Table of Contents
6. List of Figures
7. List of Tables
8. Abbreviations table
9. **Chapter 1: Introduction**
10. **Chapter 2: Existing work**
11. **Chapter 3: Methodology**
12. References
13. Appendix

Chapter 1: Introduction

1.1 Chapter Overview

This is a small introduction to your chapter.

1.2 Problem background

This is where you convince the reader why this project must be carried out, the significance of the product and the wow factor, the magnitude of the problem, etc. Use inline citations.
Detailed explanation about the problem identified, including stats to justify why it is a problem.
Include Examples in the problem and Describe your problem using example scenarios.

1.3 Problem Statement

Problem, in a nutshell just one sentence

1.4 Proposed Solution

Explanation of the proposed solution

1.5 Aim

One sentence in italics

Further, elaborate on the aim, so the reader understands and gets a clearer picture

1.6 Project Scope

Include the things you will do & won't do as part of your final product.

1.6.1 In-scope
1.6.2 Out-scope

1.7 Rich Picture Diagram

It should be very simple and easy to understand.

1.8 Resource requirements

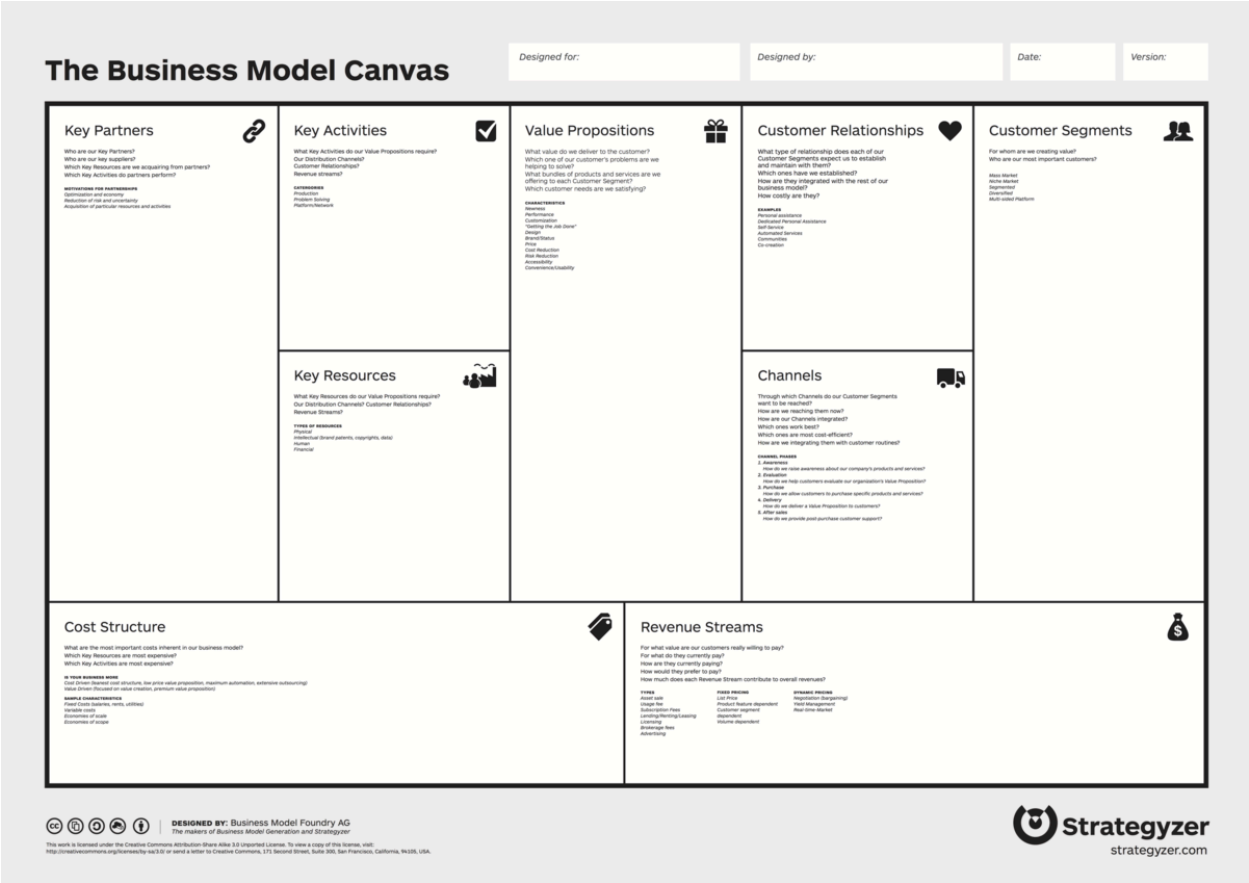
Resources needed to work on the minimum viable product

1.8.1 Hardware requirements
1.8.2 Software requirements
1.8.3 Data requirements (optional)

If you are doing a data science project, you may require appropriate datasets to work on the tasks. You need to identify them and include those requirements here.

1.9 Business model canvas

Add the following Business Model Canvas template to your report to match your project. Can be downloaded from here.



1.10 Chapter Summary

This is a small summary of your chapter.

Chapter 2: Existing work

It is up to you what sub-topics you would like to include in this chapter as every product is different from one another.

2.1 Chapter Introduction

This is a small introduction to your chapter.

2.2 Existing Work

Mention the attempted solutions of the Competitors (eg - feature comparison if there are any commercially available products). This will be helpful if you apply for competitions in the future.

Add your identified sub topics in whichever way suits your documentation in this chapter.

Topics ideally include:

- *Include a review on benchmarking or similar if applicable.*
- *This should ideally include technology/Approaches/Algorithms Review*
- *Also include a review on testing and evaluation metrics*

2.3 Tools and Implementation plan

This is where you will discuss the techniques that need to be done for your final product. For eg: if you are using a machine learning approach, you may need to talk about the different steps and the best techniques for them. Similarly you can identify the best tools for your approach here, too.

If you're building a product, you can discuss about the rough implementation plan and what you're planning to do.

2.4 Chapter Summary

This is a small summary of your chapter.

Chapter 3: Methodology

3.1. Chapter Overview

This is a small introduction to your chapter.

3.3. Development Methodology

What is the life cycle model you have selected and why?

3.4 Design methodology

Eg: What design methodology suits your project? SSADM or OOAD or Anything else?

3.5. Project Management Methodology

Critically evaluate the project management methodology you are going to use for the project

3.6 Team Work Breakdown Structure (WBS)

Mention how the work was divided among the team members and what each member has contributed on

3.7 Gantt chart diagram

Add a Gantt chart for your project (Refer sample thesis for examples)

3.8 Usage of Project Management and Collaboration Software In the project

Provide evidence
Eg -
Trello board for Project Management
Scrum
Slack for collaboration with proof screenshots
Weekly team meetings log

3.9 Risks and Mitigation

Discuss the risks you may face during the project and their mitigation plans

Eg:

Risk Item	Severity	Frequency	Mitigation Plan
	5	**5**	

	5	**4**	
	5	**1**	

3.10 Chapter Summary

This is a small summary of your chapter.

References

Harvard referencing Westminster style ONLY accepted.
REFERENCING YOUR WORK: USING WESTMINSTER HARVARD

For this the page numbering should be uppercase Roman numerals

Appendix

Each section under the appendix should be named "Appendix A", "Appendix B" etc.
For this the page numbering should be Uppercase roman numerals

Section 2 - Topics to be covered in the individual report

The following topics must be covered in the report. (Since this report is a continuation from the group report the chapter numbering should remain and it should still have all the other parts of the professional report as noted below)

1. Cover Page
2. Declaration Page
3. Abstract + Keywords
4. Acknowledgement
5. Table of Contents
6. List of Figures
7. List of Tables
8. Abbreviations table
9. **Chapter 4: System Requirements Specification (SRS)**
10. **Chapter 5: Social, Legal, Ethical and Professional Issues**
11. **Chapter 6: System Architecture & Design**
12. References
13. Appendix

Chapter 4: System Requirements Specification (SRS)

4.1 Chapter Overview

This is a small introduction to your chapter.

4.2. Stakeholder Analysis

Identify the stakeholders for your project. Draw the Onion model diagram to depict this as well add the stakeholder descriptions for clarity.

4.2.1.Onion Model

A sample is given below, don't just use that.

4.2.2.Stakeholder Descriptions

Stakeholder	Viewpoint

Functional beneficiary	
Financial beneficiary	
Social beneficiary	
Operational beneficiary	
Negative Stakeholders	
Regulatory	
Experts	
Neighboring systems	

4.3. Selection of Requirement Elicitation Techniques/Methods

Identify appropriate techniques for your requirement elicitation tasks. Eg: Questionnaires, interviews etc.

4.4. Discussion/ Analysis of Results

For each of the elicitation methods used, discuss and analyze the outcomes.

4.5. Use Case Diagram

Identify the use cases and draw the use case diagrams.

4.6. Use Case Descriptions

Use the following template for the use case descriptions

Use Case Name		
Use Case ID	**UC-001**	
Description		
Priority		
Primary Actor		
Supporting Actors		
Pre-Conditions		
Trigger		
Main flow	**Actors**	**System**
Exception flow	**Actors**	**System**
Alternate flow	**Actors**	**System**
Exclusions		
Post Conditions		

4.7. Functional Requirements (with prioritization)

Prioritize the functional requirements for your final product. Recommend to use the below template

- Critical – The requirements that are critically needed in the successful completion
- Desirable – The requirements that can add value, but are not required immediately
- Luxury – The requirements that would add luxury to the system

Requirements list		**Priority Level**	**Description**
FR1			
FR2			

4.8. Non-Functional Requirements

All non-functional requirements need to be added here.

4.9. Chapter Summary

This is a small summary of your chapter.

**In this chapter, you may discuss diagrams that are most appropriate to your topic and the above is only an indicative. (Context diagrams, domain model diagrams etc.)*

Chapter 5: Social, Legal, Ethical and Professional Issues

5.1. Chapter Overview

This is a small introduction to your chapter.

5.2 Dataset Ethical Clearance

Clear evidence of ethical clearance for the dataset should be clearly provided. If the dataset is downloaded from any data repositories (e.g., Kaggle), you should also include proper citation along with the license information.

5.3 SLEP Issues and Mitigation

Discuss in detail using the BCS code of conduct the SLEP issues relevant to your project and how it was mitigated. You do not have to state the BCS code of conduct explicitly here, however, establish the relationship with the BCS conduct with the SLEP.

5.4 Chapter Summary

This is a small summary of your chapter.

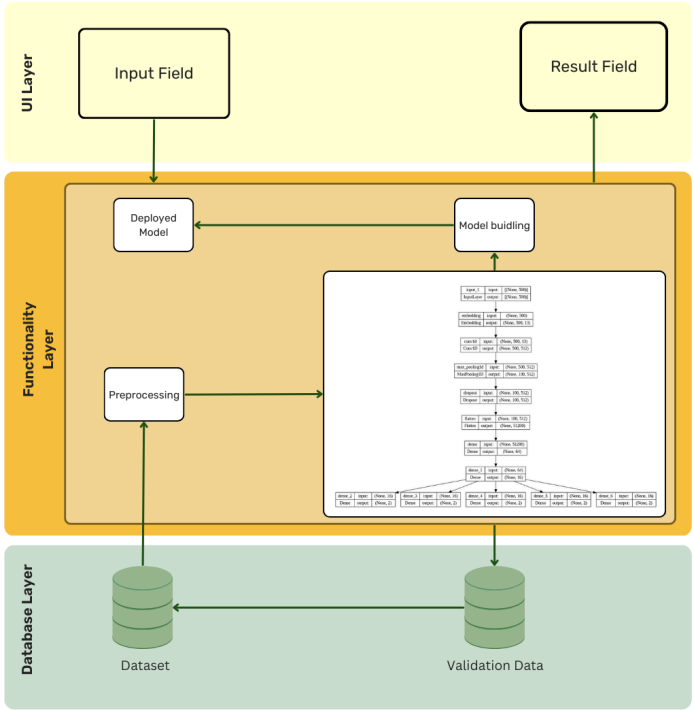
Chapter 6: System Architecture & Design

6.1. Chapter Overview

This is a small introduction to your chapter.

6.2. System Architecture Design

Depending on your topic add the appropriate architecture diagram.(Eg: Layered Architecture/Tiered Architecture/etc.) A sample is given below don't just copy paste it.



6.3. System Design

Depending on the system, add appropriate design diagrams such as follows:

6.3.1. Class Diagram/Data Flow Diagram (DFD)

6.3.2. Sequence Diagram

6.3.3. UI Design and mockups – des/high fidelity prototype (This can be done as a group and everyone can contribute to this)

6.3.4. Activity Diagram

6.4. Chapter Summary

This is a small summary of your chapter.

References

Harvard referencing Westminster style ONLY accepted.
REFERENCING YOUR WORK: USING WESTMINSTER HARVARD

For this the page numbering should be uppercase Roman numerals

Appendix

Each section under the appendix should be named "Appendix A", "Appendix B" etc.
For this the page numbering should be Uppercase roman numerals